MASTER'S DEGREE IN COMPUTATIONAL FLUID DYNAMICS · UPV

Thermal Analysis and CFD Simulation of a Small Satellite Cooling System

Active single-phase liquid cooling loop · Conjugate heat transfer · LEO sunlit orbital phase

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Why thermal control matters for CubeSats

CubeSats are cost-effective for LEO missions, but their compact form factor creates a hard thermal problem.

The limited external surface must reject internal electronic heat and absorbed solar irradiation, while keeping every component inside strict temperature limits.

This work models an active liquid-cooling loop with Simcenter STAR-CCM+, resolving conduction, convection and radiation simultaneously during the sunlit orbital phase.



1.0 W

Internal electronics heat load



1360 W/m²

LEO solar constant (Gsolar)



3.0 K

Deep-space radiative sink



−20…+60 °C

Safe operating temperature range

Project Objectives

01

Thermal Validation

Verify the electronics board stays within −20°C to +60°C under combined internal (1 W) and solar loading (408 W/m²).

02

Fluid Characterization

Characterize the velocity distribution inside the closed cooling loop and assess convective heat extraction efficiency.

03

Solar Asymmetry Analysis

Evaluate the asymmetric thermal response of the external aluminum chassis to the directional solar input.

Geometry: CubeSat thermal path

Aluminum frame

Primary conduction path for the whole assembly.

Acrylic insulation

On three faces — channels heat toward the active cooling path.

Perimeter cooling loop

Single-phase liquid driven by a virtual pump, coupled via 4 mounting brackets.

Electronics board

Central solid plate: 1.0 W uniform volumetric heat source representing PCBs and processing units.

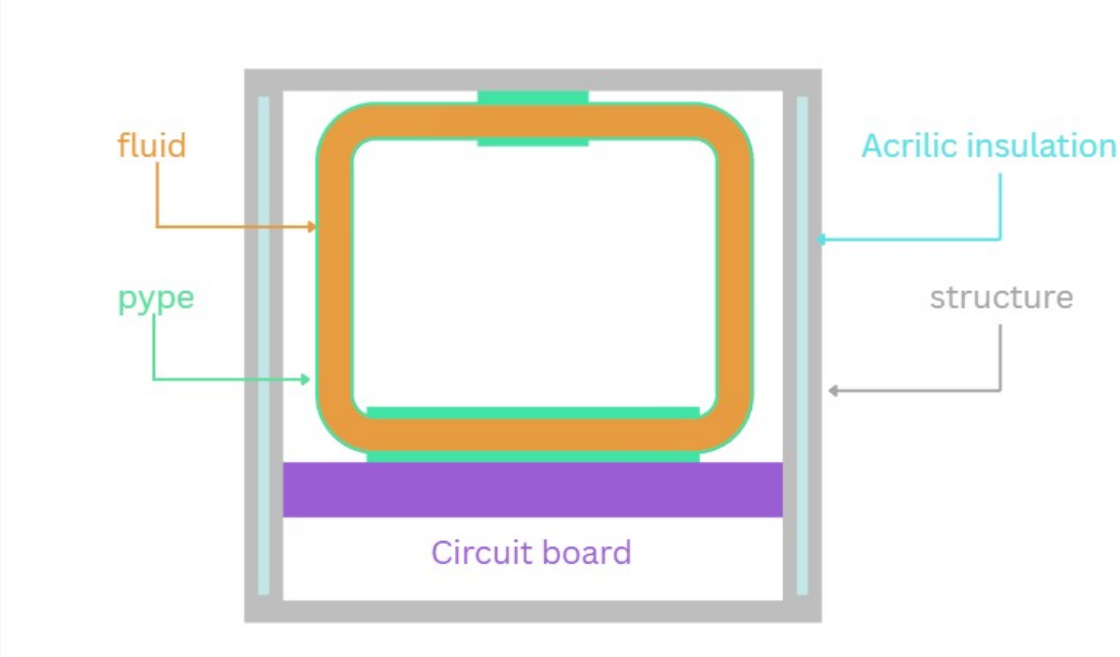
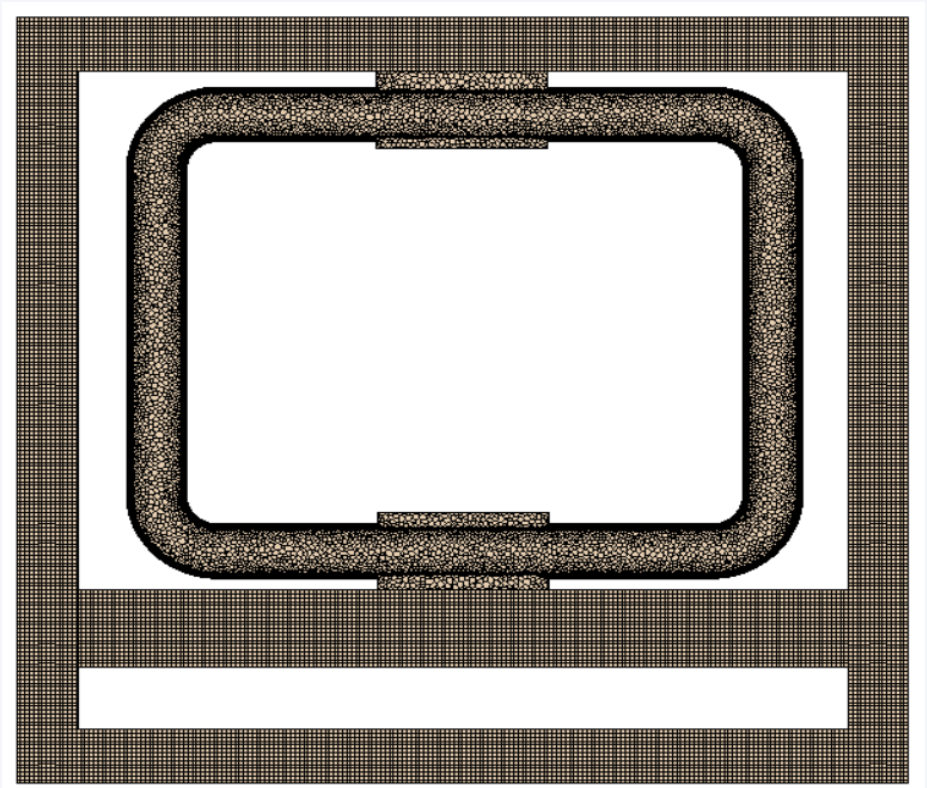


Fig. — CubeSat cross-sectional view in STAR-CCM+.

Computational mesh: hybrid strategy



Trimmed Cell Mesher (solids) + Polyhedral/Prism Layer pipeline (cooling loop).

Structural solids

Trimmed Cell Mesher + Surface Remesher, 5×10^{-4} m baseline — ideal for orthogonal geometries.

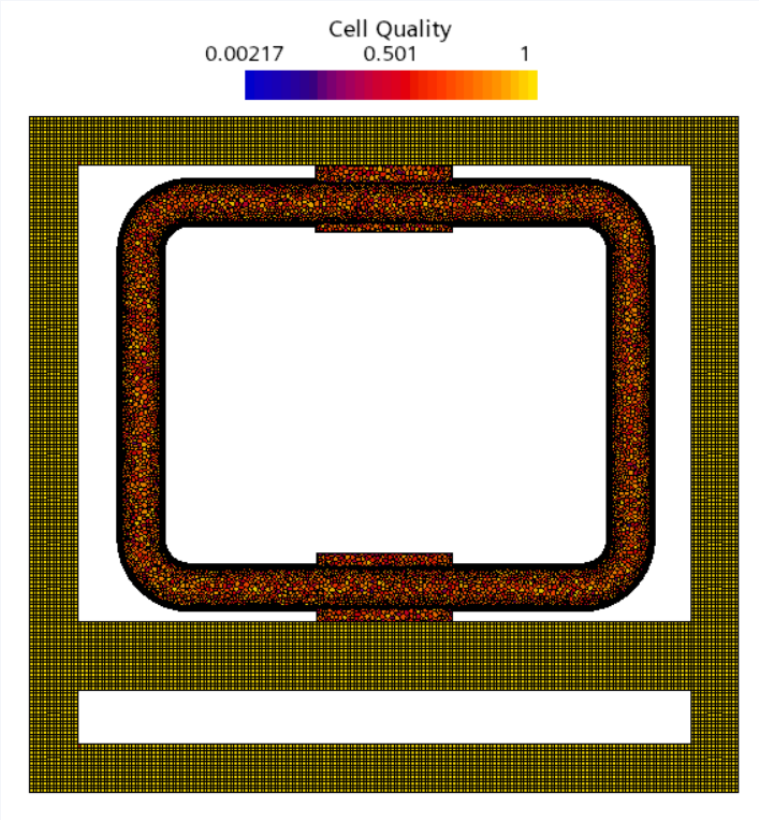
External pipe wall

Thin Mesher (5 thin layers) + Polyhedral Mesher, capturing the extremely small wall thickness.

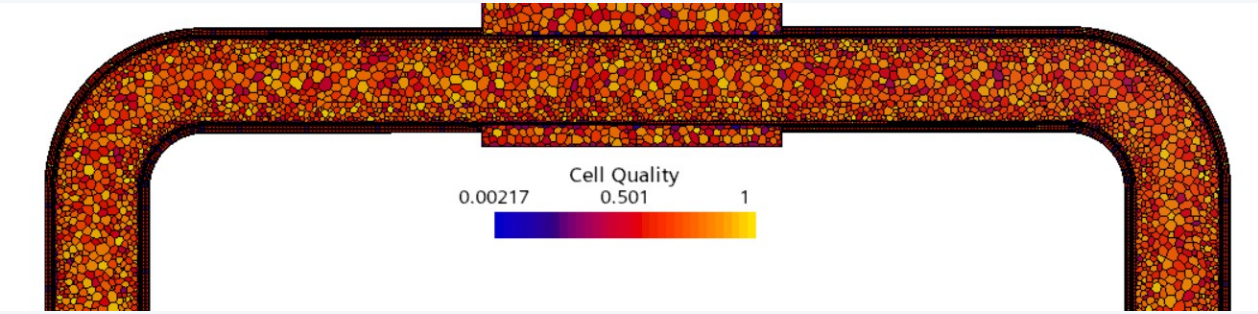
Internal fluid domain

Prism Layer Mesher to resolve hydrodynamic & thermal boundary layers, with a Polyhedral core (Generalized Cylinder Mesher).

Mesh quality validation



Global Cell Quality distribution — vast majority close to 1.0.



Detail: Prism Layers at fluid-solid wall (near-zero quality gradient) + polyhedral core.

Minimum cell quality well above critical thresholds throughout the domain — a well-conditioned grid for solver convergence.

Physical models & space thermal environment

Fluid domain

Steady laminar single-phase liquid. Coupled Fluid Energy solver for convective heat absorption.

Solid domain

Steady-state conduction through aluminum. 1.0 W uniform volumetric source on the board.

Space environment

No convection in LEO. Pure radiation to deep-space sink at Tspace = 3.0 K.

Radiation boundary condition — sun-exposed panel:

$$q_net = \alpha \cdot G_solar - \sigma \cdot \varepsilon \cdot (T^4 - T_space^4)$$

α = 0.30 (solar absorptivity) · ε = 0.85 (IR emissivity) · σ = 5.67×10⁻⁸ W/m²K⁴

q_solar = 0.30 × 1360 W/m² = 408 W/m² on one exposed face · T_space = 3.0 K

Solution method & virtual pump modeling

Solver formulation

Steady-state, pressure-based segregated solver — SIMPLE pressure-velocity coupling.

Second-order upwind schemes for momentum and energy convective terms.

Incompressible liquid model (Ma ≪ 0.3).

Radiation via custom Stefan-Boltzmann User Field Functions — mathematically equivalent to S2S for uniform background sink, at far lower cost.

Virtual pump modeling

No rotating impeller geometry — the pump is a localized volumetric momentum source aligned with the local flow direction.

$$S_m = (\vec{v} \,/\, |\vec{v}|) \cdot 200.0 \;\; [N/m^3]$$

Velocity unit vector aligns the source with local flow — prevents backflow at 90° bends.

Bypasses sliding meshes or MRF — significantly lower computational overhead.

Mesh independence study

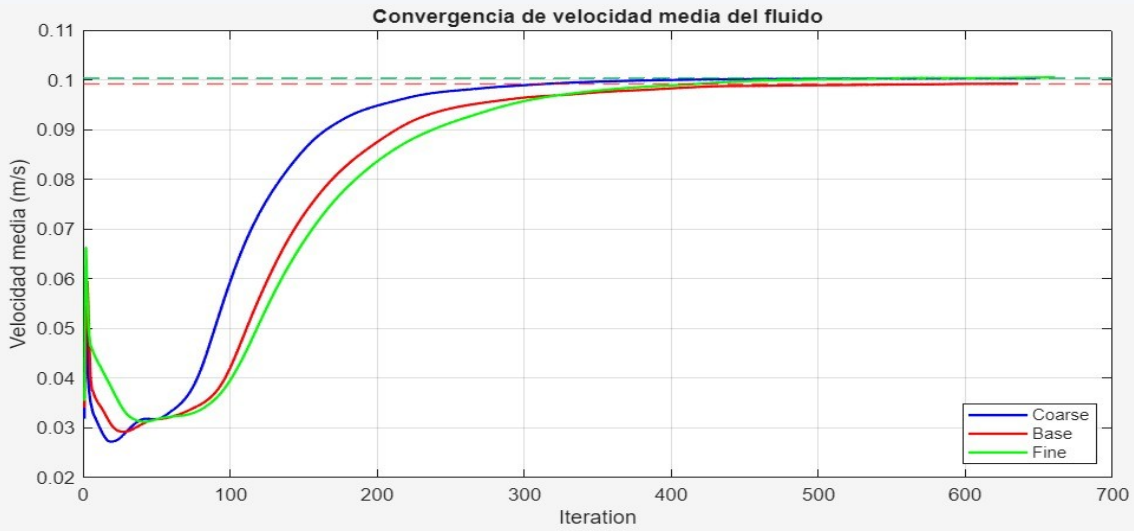
Configuration	Total cells	Board T [K]	Fluid V [m/s]
Coarse	1,034,055	303.345	0.1003
Medium (selected)	**2,610,128**	**303.341**	**0.0992**
Fine	5,799,488	303.232	0.1004

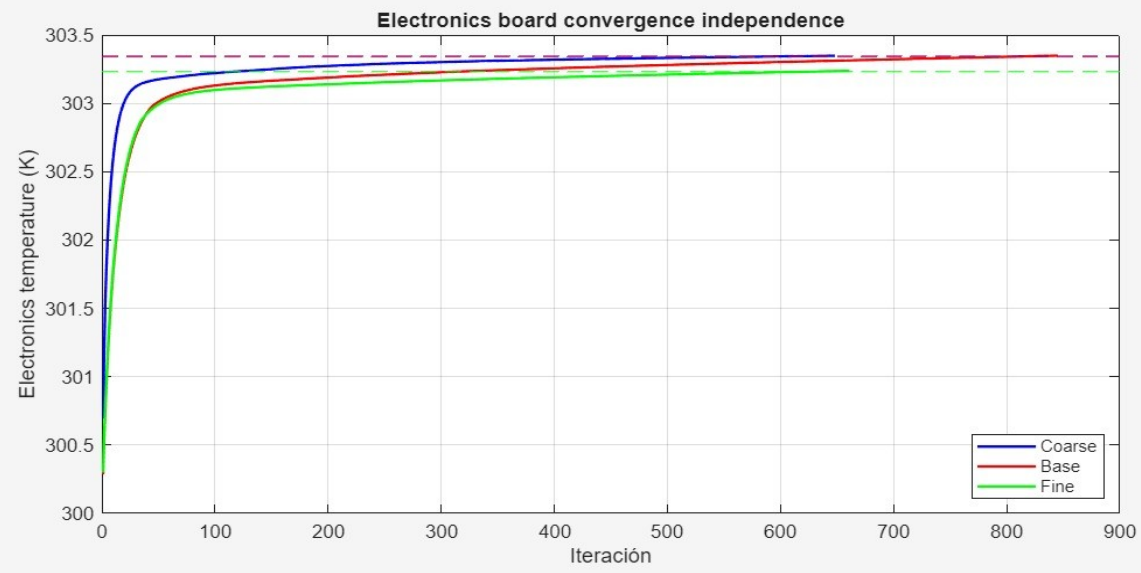
Medium mesh selected — production grid

Velocity variation (Medium → Fine): **≈ 1.23%**

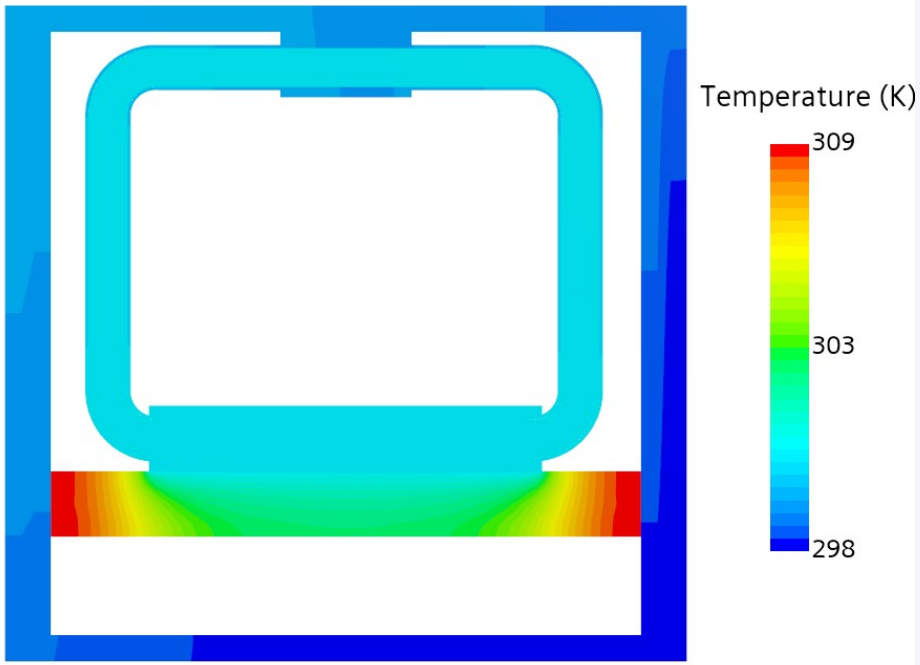
Temperature discrepancy: **0.11 K (0.036%)**

Grid-independent solution achieved at reduced computational cost.





Global thermal equilibrium & solar asymmetry



Cross-sectional temperature field (fluid domain, 300–301 K).

~309 K

Peak temperature at electronics board lateral extremities

~303–304 K

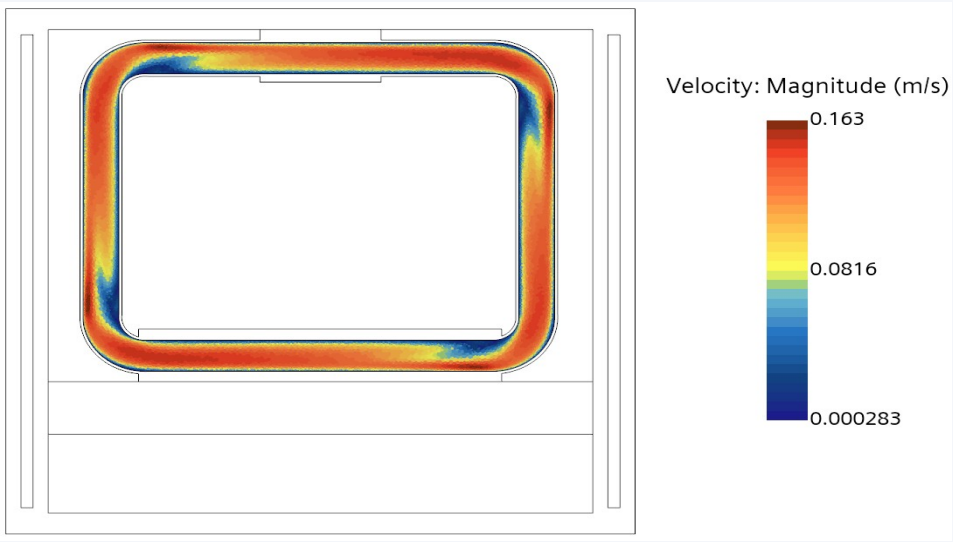
Board temperature at the cooling-channel interface

299.5 K vs 298 K

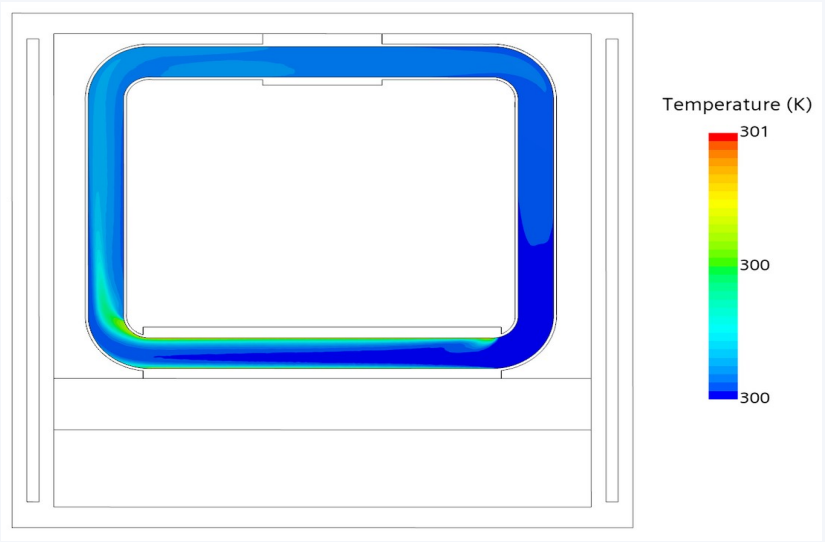
Solar-exposed panel vs shadowed panel — 1.5 K chassis gradient

The internal fluid loop acts as a dominant thermal sink, establishing a strong conductive gradient driving heat from the board toward the coolant — while the directional solar flux conducts asymmetrically across the aluminum frame.

Internal fluid dynamics & thermal core behavior



Velocity magnitude (0.000283 – 0.163 m/s).



Fluid temperature distribution (300 – 301 K).

0.12–0.14 m/s

Baseline velocity in straight channels

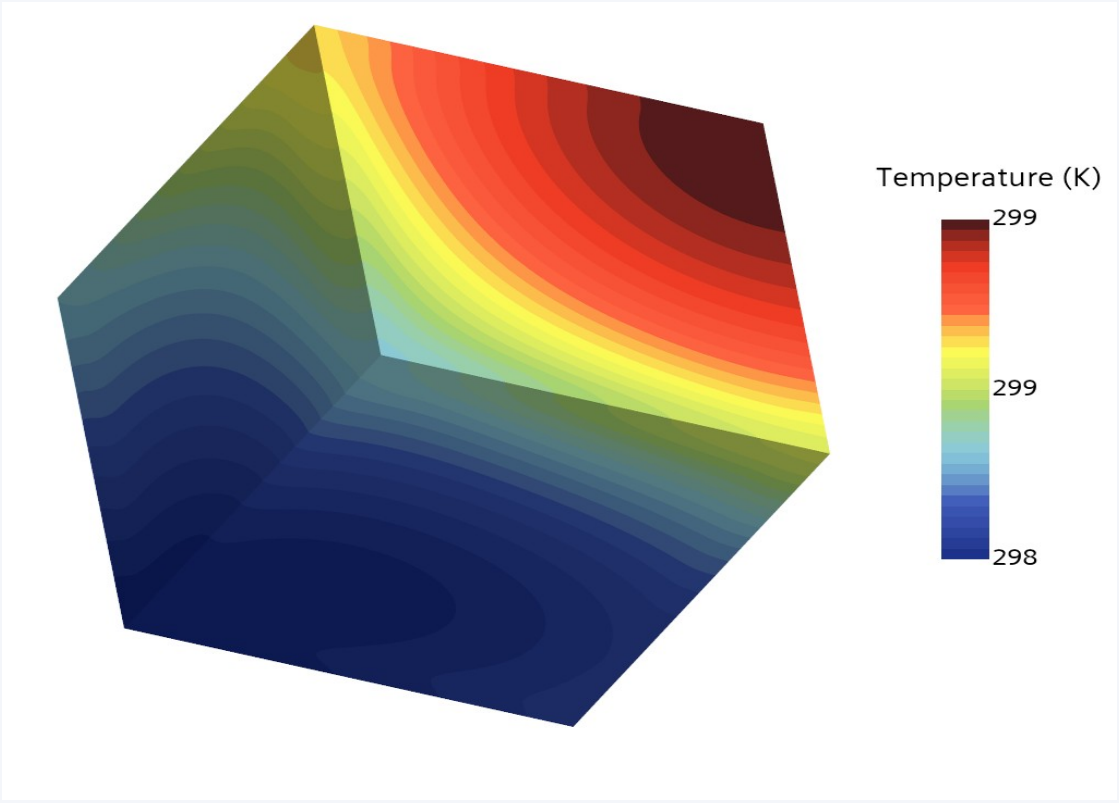
0.163 m/s

Local peak at 90° bend outer radii

< 1.0 K

Total coolant temperature rise

External chasis temperature & radiative dissipation



External chassis temperature field — 298 K (shadowed) to 299.5 K (sun-exposed face).

~299.5 K

Sun-exposed face — 408 W/m² absorbed solar flux.

~298.0 K

Shadowed panels, settling at the colder baseline.

≈ 1.5 K total gradient

Aluminum conduction redistributes the directional solar load before the T^4 radiation boundary condition rejects it to the 3.0 K deep-space sink.

Conclusions

Thermal Validation

Board stabilizes at ~303.34 K (~30.19°C) — comfortable ~30 K margin below the +60°C upper limit.

Fluid-Dynamic Efficiency

Stable average velocity of 0.099 m/s; coolant temperature rise < 1.0 K — no thermal saturation.

Radiative Balance

~1.5 K chassis gradient between exposed/shadowed panels; stable global equilibrium with deep space.

Design Insight

Near-zero stagnation at bend inner radii — avoid brackets there; consider rounded corners or guide vanes.

Future work: transient orbital cycling (sunlit/eclipse phases) and higher-emissivity surface coatings.

Summary on Choice of Project and Methodology

Project Selection

CubeSat's high power to volume ratio and constrained radiating area make it a well-bounded, industrially relevant thermal design challenge.

The sun exposed phase represents the worst-case operational condition.

Geometry and Radiation

Omitting solar panels and consolidating electronics into one heat source preserves the physically relevant thermal pathways.

Custom Stefan-Boltzmann User Field Functions replace S2S/DOM is mathematically equivalent for a uniform deep-space sink, with lower cost and explicit control of emissivity per face.

Pump Modelling

A velocity-aligned volumetric momentum source avoids rotating reference frames and sliding meshes.

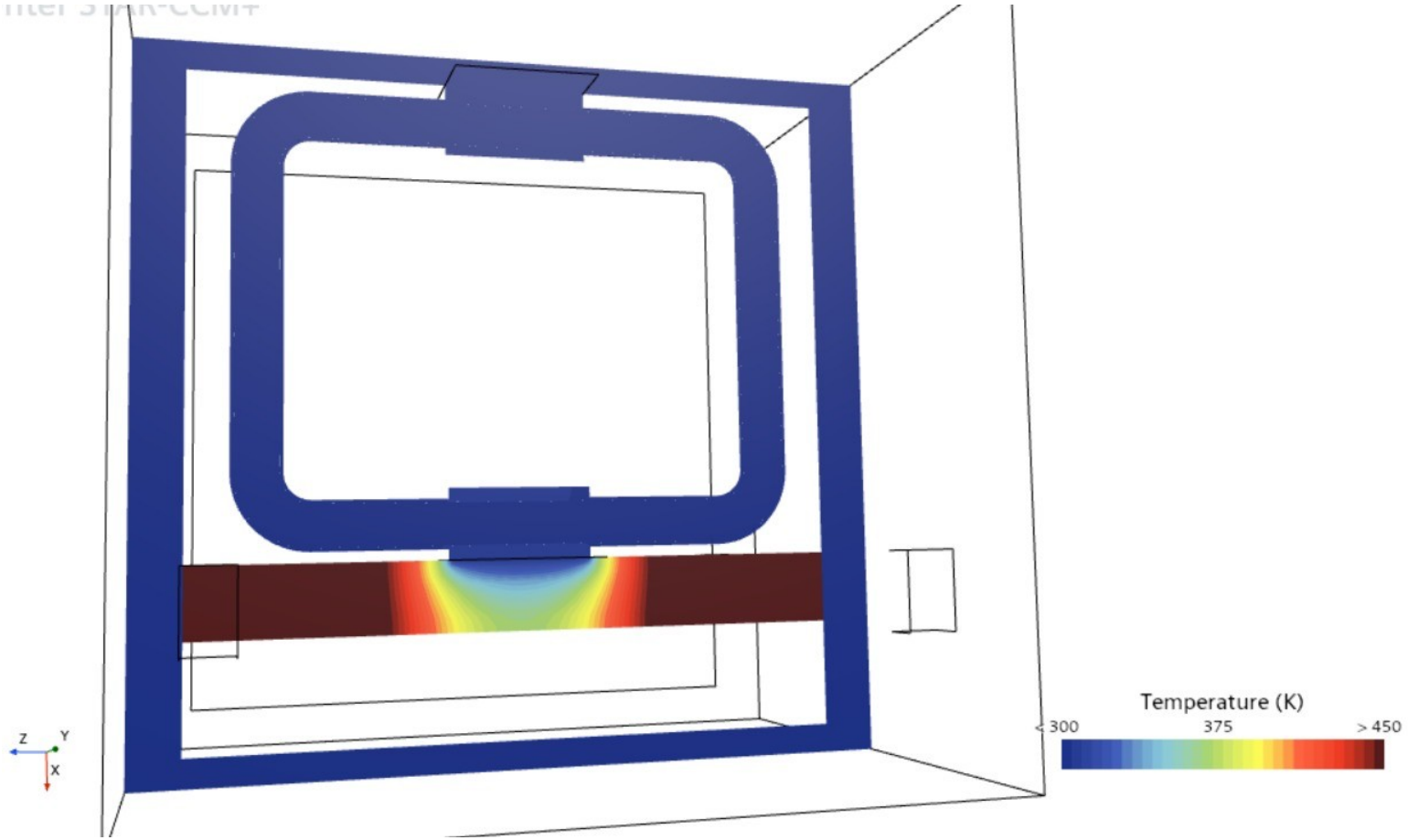
The velocity normalised direction vector prevents spurious transverse body forces.
This is a standard practice for steady-state cooling-loop simulations at this level of modelling detail.

Mesh Strategy

Prism Layers resolve the thin thermal boundary layer at the pipe wall.

Thin Mesher handles the pipe shell geometry;
Trimmed Cell Mesher aligns cells along conduction paths.
The independence study across three grids confirms the Medium mesh is free of spatial discretisation error (<1.25% variation vs Fine).

Summary on Choice of Project and Methodology





THANK YOU FOR YOU ATTENTION